JLE with auxiliary surveillance: transmission recovery under 16 surveillance settings

Generating transmission: R_{eff}=0.50, k=0.15; 5000 observed clusters and 5000 independent auxiliary trials per unique setting

Joint 2-D 95% profile contour | JLE estimate | True transmission pair | Separate 1-D 95% profile CIs

(a) p_p=0.25, p_a=1.00 | (b) p_p=0.50, p_a=1.00 | (c) p_p=0.75, p_a=1.00 | (d) p_p=1.00, p_a=1.00

(e) p_p=0.25, p_a=0.75 | (f) p_p=0.50, p_a=0.75 | (g) p_p=0.75, p_a=0.75 | (h) p_p=1.00, p_a=0.75

(i) p_p=0.25, p_a=0.50 | (j) p_p=0.50, p_a=0.50 | (k) p_p=0.75, p_a=0.50 | (l) p_p=1.00, p_a=0.50

(m) p_p=0.25, p_a=0.25 | (n) p_p=0.50, p_a=0.25 | (o) p_p=0.75, p_a=0.25 | (p) p_p=1.00, p_a=0.25

Effective reproduction number, R_{eff}

0.6
0.5
0.4

0.05 0.1 0.15 0.2 0.25 0.3 0.35

Dispersion parameter, k (linear scale)